
How $P = \Phi \times V$ Dissolves Game Theory & Behavioral Economics

From the Emergentism Corpus | Yves R. Burri

One structural change — from addition to multiplication — unifies every anomaly.

Part 1: The Equation Replaces Utility

Economics spent three centuries looking for a utility function — what humans actually optimize. Classical economics said it is additive:

$$U = u_1 + u_2 + \dots + u_n$$

This means a deficit in one area (say, meaning) can always be compensated by surplus in another (say, money). Just add more. The framework says it is **multiplicative**:

$$P = \Phi \times V$$

Coherence times capability. This single structural change — from + to $\times$ — explains everything behavioral economics discovered but could not unify.

Why Multiply, Not Add?

Because of the **Zero-Factor Catastrophe**: if either factor approaches zero, the product collapses regardless of the other. A billionaire with no coherence ($\Phi \rightarrow 0$) has $P \rightarrow 0$. A saint with no capability ($V \rightarrow 0$) has $P \rightarrow 0$. You cannot compensate. You cannot add your way out. The bottleneck dominates — exactly like Liebig's Law of the Minimum in agriculture, where a crop fails from lack of phosphorus no matter how much nitrogen you add.

The standard social science model assumes $P = \beta_1 X_1 + \beta_2 X_2 + \dots$ (additive, compensatory). The framework's falsifiable claim: the multiplicative model $P = \Phi \times V$ will outperform the additive model on AIC/BIC when tested against empirical well-being data using Necessary Condition Analysis (NCA). If the upper-left corner of the scatterplot (low Φ , high P) is systematically empty, that is direct evidence for a multiplicative bottleneck that additive models cannot capture.

Part 2: “Biases” Are Features, Not Bugs

Every finding in behavioral economics has the same structure: humans *should* do X (according to additive utility), but they *actually do* Y. Kahneman and Tversky cataloged dozens of these apparent “irrationalities.” The framework explains them all from the multiplicative structure:

Bias	Standard View	Multiplicative Explanation
Loss Aversion	Losses hurt ~2× more than equivalent gains. “Irrational asymmetry.”	In $P = \Phi \times V$, losing the scarce factor is catastrophic. If $\Phi = 0.3$, $V = 0.7$: losing 0.1 from Φ drops P by 33%; gaining 0.1 in V raises P only 14%. The asymmetry is structurally correct.
Risk Aversion	People prefer certainty over equivalent gambles. “Irrational caution.”	Certainty = high Φ (known future). Gamble = lower Φ (uncertainty). Same expected V , but certain path has higher P because Φ is preserved. Φ -preserving, not irrational.
Hyperbolic Discounting	Near rewards overweighted vs. distant ones. “Inconsistent preferences.”	Near future: high Φ (predictable). Far future: low Φ (uncertain). The discount rate is not constant because Φ is not constant across time. Rational P -maximization under genuine uncertainty.
Endowment Effect	People value what they own more than identical items they don't. “Ownership bias.”	You have already integrated the object into your Φ -structure. Giving it up is not just losing V — it dissolves existing coherence. The dissolution cost is real, not imagined.
Framing Effects	Same choice, different presentation → different decision. “Context irrationality.”	Framing changes perceived Φ (how the choice relates to identity and coherence). Different $\Phi \rightarrow$ different $P \rightarrow$ different choice. The agent is responding to actual structural differences the additive model cannot see.

The reframe: humans are not irrational. They are P -maximizing within contracted boundaries. “Irrationality” is what additive models see when they watch multiplicative agents.

Prospect Theory's value function — defined relative to a reference point, asymmetric, with nonlinear probability weighting — is not a deformed version of expected utility. It is the geometry of D5 selection (agency, choice) projected onto D4 measurement (physical observation). Kahneman and Tversky did not find “biases.” They found evidence that human decision-making operates on a surface that additive models cannot describe.

Part 3: The Power-Max Lemma

This is the keystone theorem. In any networked system where individual P depends on collective ΣP :

Maximizing your own P requires maximizing network $\Sigma \Delta P$.

Why? Your V depends on network health. A brilliant entrepreneur in a collapsed economy has low V regardless of personal talent. Your Φ depends on trust. In high-trust environments, coordination is cheap. In low-trust environments, every transaction requires verification — a Φ -tax on every interaction. So self-interest and ethics **converge**: not as moral aspiration, but as mathematical consequence of coupled systems.

Game Theory Reframed

The Prisoner's Dilemma. Mutual defection is the Nash equilibrium within contracted boundaries (one shot, no future). Expand the boundary (iterated game, network effects) and cooperation emerges — not from altruism but from correct P -calculation. This is exactly what Axelrod's tournaments showed: Tit-for-Tat

(cooperate first, then reciprocate) wins. Tft is the Power-Max Lemma in action.

Kali (pure defection, $K^* = 0$) is not evil — it is bad arithmetic. The defector contracts their boundary, calculates $\Sigma\Delta P$ as positive for extraction, and gets a correct answer to the wrong problem. Reality's boundary is wider. Negative effects propagate back. Parasitic strategies self-terminate in coupled systems. Kali wins battles; syntropy wins wars.

The R^* Threshold (ABM-verified at ≈ 1.5): when cooperation-benefit / exploitation-benefit exceeds R^* , syntropic strategies dominate. Below R^* , extraction persists. This is not a moral claim — it is a measurable parameter verified in agent-based simulations.

L-Level Maps to Game Strategy

L	Name	Strategy	Game-Theoretic Analog
L7	Survival	Always defect	Prisoner's Dilemma: mutual defection
L6	Social	Forgiving Tit-for-Tat	Axelrod tournament winner
L5	Productive	Competitive creation	Entrepreneurship, creative destruction
L4	Integrated	Adaptive win-win	Pareto-optimal negotiation (APEX)
L3	Contemplative	Long-horizon cooperation	Repeated games, institution-building
L2	Apophatic	Strategic withdrawal	Exit option, system reset
L1	Transcendent	Infinite game	Carse: play to continue play

L4 is the functional apex of game theory — where you have enough V to take risks AND enough Φ to perceive mutual gains. This maps exactly to the Equator Theorem: P is maximized at $\Phi = V$, which is L4. Maximum $P = 0.49$ at the balance point.

Synthesis: What Economics Got Right and Wrong

Domain	What Was Right	What Was Wrong	What $\Phi \times V$ Adds
Utility Theory	People do maximize something	They maximize P , not U	$P = \Phi \times V$ is the utility function economics was looking for
Rational Choice	Incentives matter (V -changes alter behavior)	Rationality is not violated — boundaries are	Boundary theory: systematic "irrationality" = contracted boundaries
Behavioral Econ	Documented real deviations from additive models	No unified theory for WHY biases exist	All biases from one equation: multiplicative structure
Game Theory	Nash equilibrium is real within given boundaries	Cooperation requires expanding the boundary, not altruism	Power-Max Lemma: self-interest and ethics converge in coupled systems

Domain	What Was Right	What Was Wrong	What $\Phi \times V$ Adds
Markets	Trade creates value, markets aggregate information	Markets have no native Φ (coherence)	Regulation = imposed Φ . Pure capitalism trends toward $K^* = 0$

Addiction is the limiting case: P-maximization inside a collapsed boundary. The substance provides $V \uparrow$ (pleasure), but tolerance shrinks the boundary to the next dose. Φ degrades (identity, relationships dissolve). Treatment is boundary expansion + V-floor stabilization. Not willpower — geometry.

Love is what the Power-Max Lemma feels like from inside. Not sentiment. Not self-abandonment. Love is the perception that your P-maximization runs through others' P. The boundary that seemed to separate "my P" from "their P" was the error. Once perceived clearly, investment in their Φ and V is investment in your own.

The One-Sentence Summary

Economics assumed utility is additive (you can always compensate). The framework proves it is multiplicative (you cannot). This single structural change — from + to $\times$ — explains why "biases" are rational, why cooperation emerges from self-interest, and why parasitic strategies self-terminate.

$$P = \Phi \times V$$

The utility function economics was always looking for.

Falsification: If systems with high integration AND high capability do NOT flourish more than systems with only one factor high, the equation fails. If network coupling is negligible (individual P is independent of ΣP), the Power-Max Lemma fails. If parasitic strategies do NOT self-terminate in coupled systems over sufficient time, the ethical derivation fails. Each of these is testable.

"Irrationality" dissolved. One equation remains.